IMPROVED KISSING NUMBERS IN NINETEEN AND TWENTY-ONE DIMENSIONS

ABSTRACT. We prove that the kissing numbers in 19 and 21 dimensions are at least 12276 and 31560. The best previously published lower bounds were 11948, due to Ho, and 29768, due to Cohn and Li. Both constructions start from the odd-sign kissing configurations of Cohn and Li and add weighted sign vectors. In 19 dimensions we cut the 20-dimensional configuration by a hyperplane, delete 184 of the points in it, and add 1792 points, the orbits of 14 integer vectors under a group of 128 sign changes; each is the projection of a 20-dimensional sign vector with reweighted head coordinates and a rescaled three-dimensional tail. In 21 dimensions we keep all 27720 points of Cohn and Li and add 3840 sign vectors of words of the punctured Golay code, with weight 2 on one line of the projective plane of order 4 formed by its words of weight 5 and weight 3 elsewhere; a linear map to $\mathbb{F}_2^4$ chooses the line, and the proof is by hand. The 19-dimensional proof reduces, by the symmetry, to an explicit list of integer inequalities about the printed vectors. An independent exact program checks all pairs of both configurations.

1. INTRODUCTION

The kissing problem in $\mathbb{R}^n$ asks how many unit spheres can touch a central unit sphere without overlapping [3, Ch. 1, §2.2]. In terms of the points of tangency, it asks for the largest set X of unit vectors in $\mathbb{R}^n$ with $\langle x, y\rangle \le 1/2$ for all distinct $x, y \in X$. We write τ_n for this maximum, as in [3, Ch. 13]. It is known only for $n \in \{1, 2, 3, 4, 8, 24\}$; see [2, Table 1.1] for references.

For $17 \le n \le 23$ the best constructions were Leech's cross sections of the Leech lattice minimal vectors [5] until Cohn and Li [2] improved dimensions 17 through 21. In dimensions 19 to 21 their construction is as follows. Take the classical configuration built from a constant-weight code [6], and put an odd number of minus signs on each codeword instead of an even number. Every sign vector of a code orthogonal to the constant-weight code then has inner product at most 6 with the old points. The kissing condition allows $\frac{1}{2}\sqrt{8n}$, since the sign vector has squared norm n and the old points 8, and this is larger than 6 for $n \ge 19$. So these sign vectors can be added, one per word of a code with minimum distance at least $n/4$. Cohn and Li obtained $\tau_{19} \ge 11692$ and $\tau_{21} \ge 29768$ in this way. Ho [4] found a larger code of 1280 words for dimension 19, reaching $\tau_{19} \ge 11948$.

Theorem 1.1. *We have* $\tau_{19} \ge 12276$ *and* $\tau_{21} \ge 31560$.

Table 1 compares these values with the best published lower bounds and the known upper bounds. The gains over the previous lower bounds are 328 in dimension 19 and 1792 in dimension 21.

The mechanism is the slack between 6 and $\frac{1}{2}\sqrt{8n}$. Cohn and Li's added points are flat: each is a sign vector scaled to squared norm 8, and none of them touches an old point. We use this slack by changing the shape of the added points.

- In 19 dimensions (Section 3) the added points come from 20 dimensions. We cut the 20-dimensional configuration by the hyperplane orthogonal to the indicator vector of a set T of four coordinates. After an isometry, a point of that hyperplane has 16 *head*

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TABLE 1. Kissing number bounds in dimensions 19 and 21, as of October 2026 [1]. The previous lower bounds are the best published ones. The upper bounds are the floors of numerical semidefinite programming values [7] and are not interval-certified.

dimension	previous lower bound	new lower bound	upper bound
19	11948 [4]	**12276**	24417 [7]
21	29768 [2]	**31560**	53524 [7]

coordinates, those outside T, and a three-dimensional *tail*. Projecting a sign vector into the hyperplane gives a *flat* point: head entries ± 1 and a short tail. Our added points keep the head signs of a sign vector but reweight the head entries and rescale the tail according to its direction. This lets two added points differ in only four head coordinates when their tails differ. A code of minimum distance at least 5, such as Ho's, must avoid such pairs. We add 1792 points where Ho adds 1280. The long tails come too close to 192 old points, and we delete 184 of them. The other eight form one orbit of a group of 128 sign changes that maps the whole configuration to itself. The net gain over Ho is $1792 - 184 - 1280 = 328$.

- In 21 dimensions (Section 4) the sign vectors come from all of the punctured Golay code D, whose words of weight 5 are the 21 lines of a projective plane of order 4. Two flat sign vectors whose words differ by a line conflict. Each added point has entries ± 2 on one line and ± 3 elsewhere. Then two points whose words differ by a line P are compatible exactly when their two lines and P are three distinct lines. A linear map $D \to \mathbb{F}_2^4$ assigns the lines so that this always holds, for the 3840 words outside its kernel. We use 3840 of the 4096 words of D, against Cohn and Li's 2048, and delete nothing. The gain is $3840 - 2048 = 1792$.

In 19 dimensions the weights were found by numerical optimization and rounding. The table was found by an AI agent outside this work, starting from added points of the shape described in Section 3.4, and was given in other coordinates; we state it in ours. The proof uses only the final integer vectors. In 21 dimensions the weights are exact.

The 19-dimensional proof depends on computer calculation in an essential way, unlike the proofs in [2]. The added points are integer vectors with entries up to 4616, and the theorem is a statement about this explicit table. The 14 representatives are printed in Appendix A. Using their symmetry, the proof reduces to a list of about 160 000 explicit integer inequalities, which we check by computer. The 21-dimensional proof is by hand. A second program, written independently, checks all 75 343 950 and 498 001 020 pairs of the two configurations (Section 5).

2. The odd-sign configurations

This section fixes notation, recalls the base configurations of Cohn and Li, and proves the one lemma about them that both constructions use (Lemma 2.2).

We identify a binary word with its support. For $c \in \mathbb{F}_2^n$ we write $\mathrm{wt}(c)$ for its weight, $(-1)^c \in \{\pm 1\}^n$ for its sign vector, and $\lambda \circ q$ for the coordinatewise product of two vectors. We write $\mathbf{1}$ for the all-ones vector and $\mathbf{1}_T$ for the indicator vector of a set T of coordinates.

Let $\mathcal{O}$ be a set of words of length n with weight 8 and pairwise intersections of size at most 4. The *odd configuration* $B(\mathcal{O})$ consists of the $4\binom{n}{2}$ roots $\pm 2e_i \pm 2e_j$ and, for each $o \in \mathcal{O}$, the 128 vectors with entries ± 1 on o, zeros elsewhere and an odd number of minus signs [2, §2].

All have squared norm 8. We call these 128 vectors the *octad vectors* on o, and their set the *fiber* over o.

Lemma 2.1 (Cohn and Li)**.** *Distinct $b, b' \in B(\mathcal{O})$ satisfy $\langle b, b' \rangle \le 4$.*

Proof. Two distinct roots have inner product in $\{0, \pm 4, -8\}$. A root has two nonzero coordinates, each contributing at most 2 against an octad vector, so that inner product is at most 4. Two vectors on the same octad have odd sign patterns, so they differ in an even and positive number $d \ge 2$ of places and have inner product $8 - 2d \le 4$. Two vectors on different octads share at most four coordinates. □

The next lemma is the reason for odd signs. It is the weighted form of the argument in [2, §2].

Lemma 2.2. *Let $c \in \mathbb{F}_2^n$ meet every word of $\mathcal{O}$ in an even number of coordinates, let $q = (-1)^c$, and let $\lambda \in \mathbb{R}^n_{>0}$. Let b be an octad vector on $o \in \mathcal{O}$. Then q and b disagree in an odd number of coordinates of o, and*

$$\langle \lambda \circ q, b \rangle \;\le\; \sum_{j \in o} \lambda_j - 2 \min_{j \in o} \lambda_j,$$

with equality for some b on o. In particular, for $\lambda = \mathbf{1}$ we have $\langle q, b \rangle = 6$ if q and b disagree in exactly one coordinate and $\langle q, b \rangle \le 2$ otherwise. For a root $b = \pm 2e_i \pm 2e_j$ the inner product is at most $2(\lambda_i + \lambda_j)$.

Proof. The sign vector q has an even number of minus signs on o, and b has an odd number. So they disagree in an odd number of coordinates of o, in particular in at least one, and the bound follows. Equality holds for the odd pattern that agrees with q except at one coordinate where λ is smallest. For $\lambda = \mathbf{1}$, d disagreements give $\langle q, b \rangle = 8 - 2d$ with d odd. □

Throughout we test compatibility in integers. For nonzero $x, y \in \mathbb{Z}^n$ the angle between them is at least 60° if and only if

$$\langle x, y \rangle \le 0 \quad \text{or} \quad 4\langle x, y \rangle^2 \le |x|^2 |y|^2. \tag{1}$$

We call such a pair *compatible*; the same test applies to rational vectors after scaling. If every pair from a finite set of nonzero rays is compatible, then the rays normalized to unit length form a kissing configuration. Distinct rays give distinct points because (1) excludes cosine 1. With $\lambda = \mathbf{1}$, Lemma 2.2 shows that the sign vector of a word that meets every word of $\mathcal{O}$ evenly is compatible with $B(\mathcal{O})$ when $6 \le \frac{1}{2}\sqrt{8n}$, that is, for $n \ge 18$, with contacts when $n = 18$.

3. Dimension 19

The configuration has three ingredients.

(1) A *base*: the points of Cohn and Li's 20-dimensional configuration that lie in the hyperplane orthogonal to $\mathbf{1}_T$, with 184 of them deleted, moved into $\mathbb{R}^{19}$ by an isometry φ.
(2) A group Γ of 128 sign changes that maps the base onto itself (Lemma 3.2).
(3) Fourteen integer vectors $r_1, \dots, r_{14} \in \mathbb{Z}^{19}$, printed in Appendix A. Their orbits under Γ are the 1792 *new points*.

Base pairs are compatible by Lemma 2.1. For pairs involving a new point, the symmetry reduces the check to 160 202 inner products of explicit integer vectors, which a computer verifies (Lemma 3.4). Section 3.1 sets up the codes, Section 3.2 defines the configuration, and

Section 3.3 gives the proof. Section 3.4 explains where the fourteen vectors come from; the proof does not use it.

3.1. **The codes.** Coordinates are $1, \dots, 20$. Let $E \subseteq \mathbb{F}_2^{20}$ be the span of the twelve words

$$\{1,8,9,12,16,17,18,19\},\ \{2,10,11,14,15,17,18,20\},\ \{3,7,9,13,15,16,17,19\},$$
$$\{4,7,8,10,12,15,16,19\},\ \{5,10,12,13,15,16,17,18\},\ \{6,7,8,9,10,13,16,18\},$$
$$\{1,4,7,9\},\ \{1,5,6,18,20\},\ \{1,3,12,15,20\},\ \{1,10,13,19,20\},$$
$$\{1,3,5,6,7,13,14,15,18,20\},\ \{2,4,6,7,8,13,14,16,17,18\}.$$

On the first 19 coordinates these are Ho's twelve generators [4, §2], in his order. Our coordinate 20 is Ho's coordinate 24, the last column of his extension matrix, so we have added it to the words where that column has a 1. Ho shows that the full extension to length 24 is the extended binary Golay code [4, Prop. 2]. So E is the Golay code punctured at Ho's coordinates 20 to 23, and its dual $E^\perp$ is the Golay code shortened at the same four coordinates [8]. The 130 words of weight 8 in $E^\perp$ are the *octads*. They form the shortened Steiner system $S(5,8,24)$ with $n = 20$ of [2, Table 2.1], so $B(\text{octads})$ is Cohn and Li's 17400-point configuration in $\mathbb{R}^{20}$, which we call B_{20}.

The five octads of the Golay code that contain four given coordinates partition the other twenty coordinates into five tetrads, and these six tetrads form a *sextet* [3, Ch. 10]. Puncturing at the four given coordinates turns those five octads into five words of weight 4 in E. We call these five words the *tetrads*.

The following facts are finite computations on the 4096 words of E.

Lemma 3.1. (1) *The weight enumerator of E is*

$$1 + 5z^4 + 64z^5 + 240z^6 + 320z^7 + 250z^8 + 640z^9 + 1056z^{10} + 640z^{11} + 250z^{12} + 320z^{13} + 240z^{14} + 64z^{15} + 5z^{16} + z^{20}.$$

The five words of weight 4 are pairwise disjoint and cover all 20 coordinates.

(2) *Let $T = \{1,4,7,9\}$, one of the five tetrads. Two distinct octads meet in at most 4 coordinates. The octads meet T in 0, 2 or 4 coordinates, for 30, 96 and 4 octads respectively.*

The intersections in (2) are even because $T \in E$ and the octads lie in $E^\perp$. We call the other 16 coordinates the *head* H. For $c \in E$ we write c_T and c_H for its parts on T and H, and likewise v_T and v_H for a vector $v \in \mathbb{R}^{20}$.

3.2. **The configuration.** *The isometry.* Let

$$R = \begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix},$$

acting on the coordinates $(1,4,7,9)$ of T. The rows of $R/2$ are an orthonormal basis of $\mathbf{1}_T^\perp \cap \mathbb{R}^T$, so $(R/2)^{\mathsf{T}}(R/2) = I - J/4$. The map $\varphi(v) = (v_H, Rv_T/2) \in \mathbb{R}^{16} \times \mathbb{R}^3$ therefore satisfies

$$\langle \varphi(a), \varphi(v) \rangle = \langle a, v \rangle - \tfrac{1}{4}\Big(\sum_{j \in T} a_j\Big)\Big(\sum_{j \in T} v_j\Big), \tag{2}$$

and it is an isometry from the hyperplane $\mathbf{1}_T^\perp$ onto $\mathbb{R}^{19}$. We call the first 16 coordinates of $\mathbb{R}^{19}$ the head and the last three the *tail*, and write f_1, f_2, f_3 for the standard basis of the tail $\mathbb{R}^3$.

Base. There are 10668 points of B_{20} orthogonal to $\mathbf{1}_T$; this is the size of Cohn and Li's 19-dimensional base [2, Table 2.1]. They are of five kinds:

- the 480 roots inside H;
- the 12 roots $\pm 2(e_i - e_j)$ with $i, j \in T$;
- the 3840 octad vectors on the 30 octads disjoint from T;
- the 6144 octad vectors on the 96 octads meeting T twice whose two T-entries have opposite signs;
- the 192 octad vectors on the four octads containing T with two plus and two minus signs on T.

We keep all of them except 184 of the last 192. Let $U = \{3, 6, 8, 19\}$, so that $T \cup U$ is one of the four octads containing T, and let P be the set of the eight vectors

$$p_y = y + e_1 + e_4 - e_7 - e_9,$$

where y has entries ± 1 on U, zeros elsewhere and an odd number of minus signs. The base consists of the 10476 points of the first four kinds and the eight points of P, 10484 points b in all. They enter the configuration as $\varphi(b)$. The images of the points of P are $\varphi(p_y) = (y_H, 2f_1)$; two of them have inner product 0 if they differ in all four places of U and 4 otherwise.

Sign group. For a word $g \in E$ that misses T, let S_g change the signs of the coordinates in g. It preserves $\mathbf{1}_T^\perp$, and $\varphi \circ S_g = S'_g \circ \varphi$, where S'_g changes the signs of the head coordinates in g; we write S_g for both. Let $\Gamma \subseteq E$ be the span of the seven words

$$\{2,5,10,11,13,15\}, \quad \{2,5,10,16,17,20\}, \quad \{2,5,12,14,15,17\}, \quad \{2,10,13,14,16,18\},$$
$$\{2,3,6,10,14,15\}, \quad \{2,3,8,10,11,17\}, \quad \{2,3,11,13,16,19\}.$$

It has 128 words: 0, 48 of weight 6, 30 of weight 8, 48 of weight 10, and H. Every word of Γ misses T. The words of E that miss T form a group of order 256, which contains the four tetrads other than T. The group Γ is one of the 16 subgroups of index 2 in it that contain none of these tetrads. The sign changes S_g, $g \in \Gamma$, form a group of order 128. Each S_g is orthogonal, and $S_g S_{g'} = S_{g+g'}$.

Lemma 3.2. *Every S_g, $g \in \Gamma$, maps the 10484-point base onto itself. On the 48 octad vectors of each fiber over an octad containing T that have two plus and two minus signs on T, the group $\{S_g\}$ has six orbits of size 8: the sets of points of that fiber with a given sign pattern on T. In particular P is one orbit.*

Proof. First, S_g preserves each of the five kinds of points above. A word $g \in E$ meets every octad evenly, so S_g maps roots to roots and each fiber onto itself. Since g misses T, S_g fixes the entries on T. So it maps the 10668 points of B_{20} orthogonal to $\mathbf{1}_T$ onto themselves, kind by kind.

Second, the orbits on the fifth kind. Let $T \cup u'$ be an octad containing T. Its octad vectors with a given pattern on T are the eight sign vectors on u' with a fixed parity of minus signs. S_g acts on them through $g \cap u'$, which has even size. For each of the four tetrads u', the map $g \mapsto g \cap u'$ takes all eight even subsets of u' as values on Γ; this is a finite computation. So $\{S_g\}$ acts transitively on these eight points, and they form one orbit.

Finally, P is the set with pattern $(+,+,-,-)$ on $T \cup U$, so it is one orbit, and the 184 deleted points are the other 23 such sets, a union of orbits. Hence the base is mapped onto itself. □

New points. Let $r_1, \ldots, r_{14} \in \mathbb{Z}^{19}$ be the vectors of Table 5 in Appendix A. Their entries have absolute values at most 4616, the only zero entries are in the tail, and their squared norms lie between 63 978 448 and 64 002 793. The new points are the $14 \cdot 128 = 1792$ vectors $S_g(r_i)$, $g \in \Gamma$.

Table 2 lists the blocks. The total is $10484 + 1792 = 12276$.

TABLE 2. The blocks of the 19-dimensional configuration. The base points are given by their preimages under φ. Diagonal and axis words are defined in Section 3.4.

block	size
roots in H	480
roots in T	12
octad vectors, octad disjoint from T	3840
octad vectors, octad meeting T twice	6144
the orbit P on the octad $T \cup U$	8
new points from diagonal words	1024
new points from axis words	768
total	12276

3.3. **Proof.**

Proposition 3.3. *The 10484 base points and the 1792 vectors $S_g(r_i)$ are 12276 pairwise compatible nonzero vectors spanning $\mathbb{R}^{19}$. Hence $\tau_{19} \geq 12276$.*

The symmetry reduces every pair involving a new point to a pair with r_i in first position. A base-new pair $(b, S_g r_i)$ is carried by S_g to $(S_g b, r_i)$, and $S_g b$ is a base point by Lemma 3.2. A new-new pair $(S_g r_i, S_h r_j)$ with $i \leq j$ is carried by S_g to $(r_i, S_{g+h} r_j)$. Since S_g is orthogonal, compatibility is unchanged. So it suffices to check the pairs in two finite lists:

(i) the $\binom{14}{2} \cdot 128 + 14 \cdot 127 = 13\,426$ products $\langle r_i, S_g r_j \rangle$ with $i \leq j$, excluding $i = j$, $g = 0$;
(ii) the $14 \cdot 10484 = 146\,776$ products $\langle r_i, b \rangle$ with b in the base.

Lemma 3.4. *Let $M = 50\,000$. For every product $\langle x, y \rangle$ in lists (i) and (ii), $\langle x, y \rangle \leq (\frac{1}{2} - \frac{1}{M})|x||y|$. In particular every such pair satisfies* (1).

Proof. This is a finite computation in exact integer arithmetic. For $\langle x, y \rangle > 0$ the inequality is equivalent to $(2M\langle x, y \rangle)^2 \leq (M-2)^2 |x|^2 |y|^2$, and we verified this for all 160 202 products. As an independent check, the program of Section 5, written separately and without the symmetry reduction, verified (1) for all 75 343 950 pairs of the configuration. □

Proof of Proposition 3.3. The base consists of points of B_{20} orthogonal to $\mathbf{1}_T$, so base pairs are compatible by Lemma 2.1 and (2). Pairs involving a new point are compatible by the reduction above and Lemma 3.4. The base contains all 492 roots, which span $\mathbb{R}^{19}$. □

This proves Theorem 1.1 for $n = 19$.

3.4. **Where the table comes from.** The new points are weighted projections of sign vectors. This subsection describes the pattern. It explains the counts 1024 and 768 in Table 2 and the remarks below, but the proof above does not depend on it.

Diagonal and axis words. Let $c \in E$ with $\mathrm{wt}(c_T) \in \{1, 2, 3\}$ and $q = (-1)^c$. The projected tail $t_0 = Rq_T/2$ satisfies $|t_0|^2 = 4 - \frac{1}{4}(\sum_T q_j)^2$ by (2). When $\mathrm{wt}(c_T)$ is odd, $t_0 \in \{\pm 1\}^3$; we call c *diagonal*. When $\mathrm{wt}(c_T) = 2$, $t_0 = \pm 2 f_k$; we call c an *axis* word. The 512 words with $\mathrm{wt}(c_T) \in \{0, 4\}$ project to a zero tail, so there is no tail to rescale, and the construction does not use them.

Kept words. Let K be the span of the five tetrads, a code of size 32. The 3584 words with $\mathrm{wt}(c_T) \in \{1, 2, 3\}$ fall into 112 cosets of K. Each coset $c + K$ is a 5-cube: its vertices are its 32

words, and its five directions add one tetrad each. Let $c_{\min}$ be the smallest word of the coset, read as the integer $\sum_{j\in c} 2^{j-1}$. We keep the words c for which $c + c_{\min}$ is a sum of an even number of tetrads. This is one of the two bipartition classes of each cube: 1024 diagonal and 768 axis words, 1792 in all.

Two kept words never differ by a single tetrad, and this is forced. Two words that differ by a tetrad other than T have equal T-patterns. Their points would have head distance 4 and equal tails, with cosine above $1/2$ whatever the tail length. Two words that differ by T have the same head and opposite tails. With head entries ± 1 and tail of squared length τ the cosine is $(16-\tau)/(16+\tau)$, which is above $1/2$ for the diagonal tails used here ($\tau < 2.3$), though not for the longest axis tails.

The representatives. For each new point x there is a kept word c such that x has the sign of $q = (-1)^c$ on every head coordinate and, among the 14 tail directions $\{\pm 1\}^3 \cup \{\pm 2 f_k\}$, its tail is closest in angle to t_0. Every kept word occurs exactly once; this is a finite computation. So $r_1, \dots, r_6$ come from axis words and $r_7, \dots, r_{14}$ from diagonal words, and Γ permutes the kept words. For a representative $r = (r_H, r_{\text{tail}})$ define the *tail factor*

$$\alpha(r) = \frac{|r_{\text{tail}}|}{|t_0|} \Big(\frac{1}{16} \sum_{j\in H} r_j^2 \Big)^{-1/2},$$

the tail length relative to that of the flat point with the same head size. It lies between 0.834 and 0.872 for diagonal and between 1.235 and 1.414 for axis words. Remark 3.6 explains these ranges.

The weights. We do not have a closed form for them. They were found by numerical optimization with the symmetry under Γ imposed: a penalty on pairs with cosine above $1/2$ was minimized over all 19 coordinates of the 14 representatives, 266 real variables, and the result was normalized to length about 8000 and rounded to integers. The proof does not depend on how they were found.

3.5. **Remarks.**

Remark 3.5. The smallest value of $\frac{1}{4} - \cos^2$ over pairs involving a new point is $5393/170666864 \approx 3.16 \cdot 10^{-5}$ for base-new pairs and about $2.17 \cdot 10^{-5}$ for new–new pairs, where the largest cosine is $31\,997\,485/63\,997\,753 \approx 0.4999783$, between r_3 and one of its images. No pair involving a new point has cosine within 10^{-5} of $1/2$; 1792 base-new and 3456 new–new pairs are within 10^{-4}. All 2 389 952 contacts are between base points, and 3368 of them involve a point of P.

Remark 3.6. Consider a flat new point $(q_H, \alpha t_0)$, with all head entries ± 1 and tail factor α. A diagonal point against an octad vector on an octad disjoint from T needs $36 \le 2(16 + 3\alpha^2)$, so $\alpha \ge \sqrt{2/3} \approx 0.8165$. Two axis points with the same pattern on T at head distance 6 need $4 + 4\alpha^2 \le \frac{1}{2}(16 + 4\alpha^2)$, so $\alpha \le \sqrt{2} \approx 1.4142$. A diagonal and an axis point at distance 1 on T and 4 on H, with tail factors α and β and tails making an acute angle, have inner product $8 + 2\alpha\beta$ against the bound $\frac{1}{2}\sqrt{(16 + 3\alpha^2)(16 + 4\beta^2)}$. With $\alpha = \beta$ the inner product exceeds the bound for every α in $[\sqrt{2/3}, \sqrt{2}]$ (at $\alpha = 1$ it is 10 against 9.75), so the two factors cannot be equal. The table has the diagonal factor near 0.83 and the axis factor above 1.2.

Axis factors above 1 are what the deleted points cannot tolerate. Take a point b on an octad containing T with two signs of each kind on T, and a flat axis point with $q_T = b_T$ whose sign vector disagrees with b in exactly one place of the octad. By (2) their inner product is $6 + 4(\alpha - 1)$, above the bound $\sqrt{2(16 + 4\alpha^2)}$ for α near 1.4. For the points of P these axis points are the 128 with tail along $+f_1$, each conflicting with four points of P; they form the orbit of r_6. Since Γ moves any point of P to any other, keeping one point of P forces all

eight, and the table must change in the same way in either case. In the table, r_6 has tail factor 1.235, and its entries on $3, 6, 8, 19$ are 1453 in absolute value against 1777 on the other head coordinates, a ratio of 0.82. The other axis representatives have tail factor 1.400 (tail along $-f_1$) or 1.414 (tail along $\pm f_2$, $\pm f_3$), and the diagonal ones 0.834 or 0.872 as the first tail coordinate is negative or positive. Eight of the 14 representatives have all 16 head entries equal in absolute value, and the head entries of each of the others except r_6 vary by at most 1.6%.

Remark 3.7. For each of the 184 deleted points, the largest cosine with a new point lies between 0.5286 and 0.5525. By Lemma 3.2, a point put back on any fiber forces its whole orbit of 8 under Γ, if the configuration is to stay invariant under Γ. We do not know whether a second orbit can be kept.

Remark 3.8. Among groups of sign changes S_g with $g \in E$ missing T, the group Γ is as large as possible for this table. These words g form a group of order 256, and it contains the four tetrads u' other than T. If $S_{u'}$ were in the group, a new point x and $S_{u'}x$ would both be in the configuration, with cosine $1 - 2m/|x|^2$, where m is the part of $|x|^2$ on u'; this is at most $1/2$ exactly when $m \geq |x|^2/4$. For every r_i and every such u', $m < 0.23|r_i|^2$, so the cosine is above 0.54, and the same holds for any table whose head entries are nearly equal in absolute value. A subgroup of the 256-word group that contains none of the four tetrads has index at least 2, so order at most 128, and with it the table needs at least $1792/128 = 14$ representatives.

The numbers in Remarks 3.5–3.8 were computed by computer; the fractions are exact and the decimals are rounded.

4. Dimension 21

The configuration is Cohn and Li's base B plus 3840 *line-weighted* sign vectors: for a word w of the punctured Golay code D and a line L of the projective plane formed by its weight-5 words, the vector with entries ± 2 on L and ± 3 elsewhere, signed by w. The proof has three steps. Lemma 4.2 shows that every such vector is compatible with the base. Lemma 4.3 computes the inner product of two such vectors exactly and turns compatibility into a rule about three lines, (4). A linear map $\eta\colon D \to \mathbb{F}_2^4$ then chooses the line of each word so that the rule holds (Proposition 4.4).

4.1. **The codes.** Coordinates are $1, \dots, 24$, and the construction lives on the first 21. Let $\mathcal{G}$ be the extended binary Golay code: the cyclic code of length 23 generated by $1 + t + t^5 + t^6 + t^7 + t^9 + t^{11}$, with the coefficient of t^k at coordinate $k + 1$, extended by an overall parity bit at coordinate 24 [3, Ch. 3, §2.8]. Its nonzero weights are 8, 12, 16 and 24, two octads meet in 0, 2 or 4 coordinates, and every three coordinates lie in exactly 21 octads [3, Ch. 10, §2]. Let $Q = \{22, 23, 24\}$. Let $\mathcal{O}$ be the 210 octads of $\mathcal{G}$ disjoint from Q. Then $B = B(\mathcal{O})$ is Cohn and Li's 27720-point configuration in $\mathbb{R}^{21}$.

Let D be $\mathcal{G}$ punctured to the first 21 coordinates, a $[21, 12, 5]$ code. Puncturing is injective because $\mathcal{G}$ has minimum distance $8 > 3$, so D has 4096 words. Because $\mathcal{G}$ is self-dual and the octads in $\mathcal{O}$ vanish on Q, every word of D meets every octad of $\mathcal{O}$ evenly. A word $d \in D$ comes from a unique $g \in \mathcal{G}$, and $\mathrm{wt}(d) = \mathrm{wt}(g) - |g \cap Q|$ with $|g \cap Q| \leq 3$. So the words of weight 5, 6 and 7 come from octads g with $|g \cap Q| = 3$, 2 and 1, and all other nonzero words have weight at least 8.

We call the 21 words of weight 5 in D *lines*. They are the octads through Q with Q removed. Two of these octads share the three points of Q, so they meet in exactly 4 coordinates, and two lines meet in exactly one coordinate. Five lines pass through each coordinate, so the lines

form a projective plane of order 4 on the 21 coordinates. For a line L we write $\hat{L} = L \cup Q$ for its octad.

Two sign vectors of distinct words of D at distance d have cosine $1 - 2d/21$, which exceeds $1/2$ only for $d = 5$, where it is $11/21$. Cohn and Li add 2048 sign vectors, from a linear code equivalent to a subcode of index 2 in D with minimum distance 6 [2, §2]. To use more words of D we must allow pairs of words that differ by a line.

Lemma 4.1. *Let L be a line and $d \in D$ a nonzero word, and put $a = |d \cap L|$.*

(1) *Every octad of $\mathcal{O}$ meets L in 0, 2 or 4 coordinates.*
(2) *If $\mathrm{wt}(d) = 5$, then d is a line, and $a = 5$ if $d = L$ and $a = 1$ otherwise.*
(3) *If $\mathrm{wt}(d) = 6$, then $a \le 2$. If $\mathrm{wt}(d) = 7$, then $a \le 3$.*

Proof. (1) An octad $o \in \mathcal{O}$ is disjoint from Q, so $o \cap L = o \cap \hat{L}$, and two distinct octads meet in 0, 2 or 4 coordinates. (2) is the definition of lines and the fact that two lines meet in one coordinate. (3) Let d come from g. If $\mathrm{wt}(d) = 6$, then g is an octad with $|g \cap Q| = 2$, so $g \ne \hat{L}$ and $|g \cap \hat{L}| \in \{2, 4\}$. Since $g \cap \hat{L}$ contains the two points of $g \cap Q$, $a = |g \cap \hat{L}| - 2 \le 2$. If $\mathrm{wt}(d) = 7$, the same argument with $|g \cap Q| = 1$ gives $a \le 4 - 1 = 3$. □

4.2. **Line-weighted sign vectors.** For a line L let $\lambda_L \in \{2, 3\}^{21}$ have entries 2 on L and 3 elsewhere, so $|\lambda_L|^2 = 5 \cdot 4 + 16 \cdot 9 = 164$. For $w \in D$ and a line L put

$$x(w, L) = \lambda_L \circ (-1)^w.$$

The lighter entries lower the inner product with a point whose word differs from w on L. They cost little against the base.

Lemma 4.2. *For every $w \in D$ and every line L, the point $x(w, L)$ is compatible with every point of B. The largest inner product is 18, against an octad vector, and the largest cosine is $18/\sqrt{1312} \approx 0.49694$.*

Proof. By Lemma 2.2, the inner product with an octad vector on o is at most $\sum_{j \in o} \lambda_j - 2 \min_{j \in o} \lambda_j$. By Lemma 4.1(1), o meets L in 0, 2 or 4 coordinates, and the bound is $24 - 6 = 18$, $(4 + 18) - 4 = 18$ or $(8 + 12) - 4 = 16$. Against a root the inner product is at most $2(3 + 3) = 12$. Both are compatible, since $4 \cdot 18^2 = 1296 \le 1312 = 164 \cdot 8$. □

Lemma 4.3. *Let $w \ne w'$ be words of D, let L, L' be lines, and put $d = w + w'$. If d is a line and L, L', d are not three distinct lines, then $\langle x(w, L), x(w', L') \rangle > 82 = \frac{1}{2}|\lambda_L|^2$. Otherwise the inner product is at most 82. It equals 80 when d is a line and L, L', d are three distinct lines through one coordinate, and 82 when they are three distinct lines not through one coordinate.*

Proof. Write $\sigma = (-1)^d$, $\delta = \mathrm{wt}(d)$, $a = |d \cap L|$ and $a' = |d \cap L'|$, and let e_L be the indicator vector of L. Since $\lambda_L = 3 \cdot \mathbf{1} - e_L$, the inner product is

$$\sum_j \left(9 - 3e_L - 3e_{L'} + e_L e_{L'}\right)_j \sigma_j = 9 \sum_j \sigma_j - 3 \sum_{j \in L} \sigma_j - 3 \sum_{j \in L'} \sigma_j + \sum_{j \in L \cap L'} \sigma_j.$$

The first three sums are $21 - 2\delta$, $5 - 2a$ and $5 - 2a'$. So

$$\langle x(w, L), x(w', L') \rangle = 159 - 18\delta + 6(a + a') + \sum_{j \in L \cap L'} \sigma_j. \tag{3}$$

The last sum is $5 - 2a$ if $L = L'$. If $L \ne L'$, the lines meet in one coordinate p, and the sum is -1 if $p \in d$ and 1 otherwise.

Case $\delta \ge 6$. Then d is not a line, and we show the inner product is at most 82. By Lemma 4.1, $a, a' \le a_{\max}$, where $a_{\max} = 2, 3, 5$ for $\delta = 6$, $\delta = 7$ and $\delta \ge 8$. If $L = L'$, (3) reads

TABLE 3. The 21 lines P of D and the values $\eta(P)$, written as four bits. The 15 lines marked with an asterisk are the lines $f(s)$.

line	η	line	η	line	η
$\{4,5,6,8,10\}$	1011^*	$\{2,3,6,7,15\}$	1101^*	$\{1,4,9,14,15\}$	1000^*
$\{3,9,10,11,16\}$	1111^*	$\{1,6,12,13,16\}$	0011^*	$\{7,8,9,12,17\}$	0001^*
$\{4,7,11,13,18\}$	1100^*	$\{2,10,12,14,18\}$	0101^*	$\{1,3,5,17,18\}$	1010^*
$\{2,5,9,13,19\}$	0100^*	$\{6,11,14,17,19\}$	0111^*	$\{8,15,16,18,19\}$	0001
$\{1,2,8,11,20\}$	0010^*	$\{5,7,14,16,20\}$	0011	$\{10,13,15,17,20\}$	0010
$\{3,4,12,19,20\}$	0001	$\{3,8,13,14,21\}$	1001^*	$\{5,11,12,15,21\}$	0110^*
$\{2,4,16,17,21\}$	1110^*	$\{1,7,10,19,21\}$	0011	$\{6,9,18,20,21\}$	0010

$164-18\delta+10a$. If $L\neq L'$, it is at most $160-18\delta+6(a+a')$. In both cases the inner product is at most

$$164-18\delta+12a_{\max}=80,\ 74,\ 80 \quad \text{for } \delta=6,\ 7,\ 8,$$

and smaller for $\delta>8$. All are at most 82.

Case $\delta=5$. Then d is a line P, and by Lemma 4.1(2), $a=5$ if $P=L$ and $a=1$ otherwise, and likewise for a'. We evaluate (3).

- $L=L'=P$: $a=5$, and the inner product is $164-90+50=124$.
- $L=L'\neq P$: $a=1$, and the inner product is $164-90+10=84$.
- $L\neq L'$ and $P=L$: $a=5$, $a'=1$, and $p=L\cap L'$ lies on P, so the last sum is -1. The inner product is $159-90+36-1=104$. The case $P=L'$ is the same.
- L, L', P distinct: $a=a'=1$, and the inner product is $81+\sum_{j\in\{p\}}\sigma_j$, which is 80 if $p\in P$, that is, if the three lines pass through one coordinate, and 82 otherwise.

The first three cases are exactly those where L, L', P are not three distinct lines, and there the inner product exceeds 82. In the last case it is at most 82. □

So a set of points $x(w,L(w))$, one for each w in a set $W\subseteq D$, is pairwise compatible if and only if

$$(4)\qquad w,\ w+P\in W \text{ for a line } P \quad\Longrightarrow\quad L(w),\ L(w+P),\ P \text{ are three distinct lines.}$$

In particular $w+L(w)\notin W$ for every $w\in W$, so the label of a word must point out of W. Two points on the same word are never compatible, since $\langle x(w,L),x(w,L')\rangle\geq 160$.

4.3. **The configuration.** Define a linear map $\eta\colon D\to\mathbb{F}_2^4$ by

$$\eta(w)=\big(w_3+w_4,\ \ w_2+w_{11},\ \ w_2+w_5+w_6+w_{10},\ \ w_1+w_3+w_8+w_{14}\big),$$

where $w_j\in\mathbb{F}_2$ is the jth coordinate of w. Table 3 lists the 21 lines and their values $\eta(P)$. Every line has $\eta(P)\neq 0$, and every nonzero value of $\mathbb{F}_2^4$ occurs. For each nonzero $s\in\mathbb{F}_2^4$ let $f(s)$ be the line marked with an asterisk in Table 3 that has $\eta(f(s))=s$.

Base. All of B, the 27720 points of Cohn and Li.

New points. Let $W=\{w\in D:\eta(w)\neq 0\}$. Since η is onto, $\ker\eta$ has $4096/16=256$ words, and $|W|=3840$. For $w\in W$ let $L(w)=f(\eta(w))$. The new points are the 3840 vectors $x(w,L(w))$, $w\in W$. They are integer vectors with entries ±2 on the line $L(w)$, ±3 elsewhere, and signs given by w.

The total is $27720+3840=31560$.

Proposition 4.4. *The 27720 points of B and the 3840 vectors $x(w,L(w))$, $w\in W$, are 31560 pairwise compatible nonzero vectors spanning $\mathbb{R}^{21}$. Hence $\tau_{21}\geq 31560$.*

TABLE 4. The exact all-pairs check.

dimension	points	pairs	result
19	12276	75 343 950	pass
21	31560	498 001 020	pass

Proof. Base pairs are compatible by Lemma 2.1, and base-new pairs by Lemma 4.2. For new pairs we check the line rule (4). Let w and $w' = w + P$ be in W for a line P. We need $L(w)$, $L(w')$ and P to be three distinct lines.

- $L(w) \neq L(w')$: since $\eta(P) \neq 0$, $\eta(w') = \eta(w) + \eta(P) \neq \eta(w)$, and f is injective because $\eta(f(s)) = s$.
- $L(w) \neq P$: if $L(w) = P$, then $\eta(P) = \eta(L(w)) = \eta(w)$, so $\eta(w') = 0$ and $w' \notin W$, a contradiction.
- $L(w') \neq P$: the same argument with w and w' exchanged.

By Lemma 4.3 every new pair is compatible. The base contains all 840 roots, which span $\mathbb{R}^{21}$. □

This proves Theorem 1.1 for $n = 21$. The proof uses no computer calculation beyond listing the 21 lines of Table 3 and their 84 parities.

4.4. **Remarks.**

Remark 4.5. By Lemma 4.3, two new points touch exactly when their words differ by a line P and $L(w)$, $L(w + P)$, P are three lines not through one coordinate. There are 34 048 such pairs, and 3584 further pairs of new points have inner product 80; no base-new pair touches. These counts were computed by computer.

Remark 4.6. The rule (4) bounds this method. Each $w \in W$ has the neighbor $w + L(w)$ outside W, and each word outside W is $w + P$ for at most 21 pairs (w, P). The relation $w \sim w + P$ preserves the cosets of the span of the lines, which is a $[21, 10]$ code (a finite computation), so D splits into four classes of 1024 words, and in each class at most $\lfloor 1024 \cdot 21/22 \rfloor = 977$ words lie in W. So line-weighted sign vectors with any labeling give at most $4 \cdot 977 = 3908$ new points. We do not know whether more than 3840 are possible.

5. VERIFICATION

What is proved by hand and what by computer. By hand: Lemmas 2.1 and 2.2, the symmetry reduction (Lemma 3.2 and the lists (i) and (ii) in dimension 19), and Section 4 apart from the list of lines in Table 3 and the counts in Remarks 4.5 and 4.6. By computer, in exact integers: Lemma 3.1, the coset and orbit counts of Section 3, Lemma 3.4, the counts in Remarks 4.5 and 4.6, and the all-pairs check below.

All pairs. One program, written independently of the programs that built the configurations, checks each configuration on all pairs, with no symmetry reduction and no floating point in any decision. It confirms the number of points and that every point is nonzero. For every pair of distinct points x, y it decides (1), that is $\langle x, y\rangle \leq \frac{1}{2}|x||y|$: the coordinates are integers, and when $\langle x, y\rangle > 0$ it compares $4\langle x, y\rangle^2$ with $|x|^2|y|^2$ exactly, in 128-bit integers. It also confirms that no two points have the same direction. Table 4 gives the results. Every pair passes, and in each configuration the largest cosine is exactly $\frac{1}{2}$. This check uses neither the symmetry nor the lemmas, so it confirms Propositions 3.3 and 4.4 independently.

6. Conclusion

We have shown that $\tau_{19} \geq 12276$ and $\tau_{21} \geq 31560$, improving the best published lower bounds of 11948 and 29768. In 19 dimensions two questions are open. Can a second orbit of eight deleted points be kept, under Γ or under a smaller group with a larger table? And does the table have a closed form? Remark 3.6 suggests one: outside the orbit of r_6 it is close to flat points with one tail factor for diagonal and one for axis words, up to about 5%. In 21 dimensions, can a labeling of more than 3840 words satisfy the line rule (4), up to the bound 3908 of Remark 4.6, or can other weights use more words of D? And do weights adapted to the low-weight words of a code help in other dimensions?

Use of artificial intelligence

GPT-6-Astra Pro and Claude Opus 5.5 were used to find these constructions and draft this paper.

Data availability

The coordinates and an exact checking program accompany this paper.

Appendix A. Representatives

Table 5 lists the 14 representatives of Section 3.2, in the order of the accompanying data. The axis representatives $r_1, \ldots, r_6$ have tails along $-f_1$, $-f_2$, $+f_3$, $-f_3$, $+f_2$ and $+f_1$; r_6 is the representative of Remark 3.6. Each is written as two rows, the first with the ten head coordinates 2, 3, 5, 6, 8, 10 to 14 and the second with the six head coordinates 15 to 20 and the tail.

TABLE 5. The 14 representatives r_i in dimension 19. Columns $2, \ldots, 20$ are the head coordinates, and f_1, f_2, f_3 the tail.

	2	3	5	6	8	10	11	12	13	14
	15	16	17	18	19	20	f_1	f_2	f_3	
r_1, axis	1638	1638	−1638	−1638	−1638	1638	1638	−1638	1638	1638
	−1638	1638	−1638	1638	1638	1638	−4588	0	0	
r_2, axis	−1633	1633	1633	−1633	−1633	−1633	−1633	1633	1633	1633
	−1633	1633	1633	1633	1633	1633	−102	−4616	1	
r_3, axis	1633	1633	1633	−1633	−1633	−1633	1633	1635	1633	1633
	1633	1635	−1633	1633	1633	1633	−101	0	4616	
r_4, axis	−1633	−1633	−1633	−1633	−1633	−1633	−1633	−1633	1633	1633
	1633	1633	−1633	1633	−1633	1633	−100	0	−4616	
r_5, axis	1633	−1633	−1633	−1633	−1633	−1633	1633	−1633	1633	1633
	−1633	1633	1633	1633	−1633	1633	−102	4616	1	
r_6, axis	−1777	−1453	1777	−1453	−1453	1777	−1777	1777	1777	1777
	−1777	1777	−1777	1777	−1453	1777	4203	0	0	
r_7, diagonal	−1881	−1881	1881	−1881	−1881	1881	1881	−1881	1881	1881
	1881	1881	−1881	1881	1881	1881	−1568	−1568	−1568	
r_8, diagonal	1881	−1881	1881	−1881	−1881	1881	−1881	−1881	1881	1881
	−1881	1881	1881	1881	1881	1881	−1568	1568	1568	
r_9, diagonal	−1858	−1885	−1885	−1858	−1858	−1856	1885	1856	1885	1858
	−1885	1856	−1856	1885	1885	1885	1593	−1649	1649	
r_{10}, diagonal	1881	−1881	−1881	−1881	−1881	−1881	−1881	1881	1881	1881
	1881	1881	1881	1881	1881	1881	−1568	−1568	1568	
r_{11}, diagonal	−1885	1858	1884	−1885	−1885	−1857	1859	−1857	1884	1885
	1884	1857	1857	1884	−1858	1859	1592	1650	−1650	
r_{12}, diagonal	1881	1881	1881	−1881	−1881	−1881	−1881	−1881	1881	1881
	−1881	1881	−1881	1881	−1881	1881	−1568	1568	−1568	
r_{13}, diagonal	−1859	1859	−1884	−1884	−1884	1884	1885	1857	1884	1859
	−1857	1857	1884	1857	−1858	1885	1592	−1650	−1650	
r_{14}, diagonal	1885	1885	−1885	−1858	−1858	1885	−1858	1856	1885	1885
	1856	1856	−1885	1856	−1885	1858	1593	1649	1649	

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